Software Requirements Specification

for

Study planner and partner finder system

Version 1.0 approved

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(This project was completed as part of a 3-member group in our Software Engineering course. My contributions included documentation, SRS sections, and design work.)

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Revision History

Name	Date	Reason For Changes	Version
-	-	Initial Approved Version	1.0

1.Introduction

1.1Purpose

This Software Requirements Specification (SRS) document defines the complete requirements for the *Study Planner and Partner Finder System*. It includes system behavior, functional and nonfunctional requirements, interfaces, constraints, and detailed features of the system. This SRS describes the entire product.

1.2Document Conventions

- Requirements are labeled as **REQ-X.Y**.
- Priorities: **High (H), Medium (M), Low (L)**.
- Bold text indicates key elements.
- All diagrams referenced are conceptual.

1.3Intended Audience and Reading Suggestions

- **Developers & Designers:** Focus on Sections 2, 3, 4.
- **Testers:** Focus on functional requirements in Section 4.
- **Project Managers:** Review Sections 1, 2, and 5.
- **End-users / Clients:** Overview in Sections 1 and 2.

1.4Product Scope

The system is a web-based application that helps students:

- Plan their study schedules intelligently
- Generate auto-timelines from deadlines
- Track academic progress with analytics Find
- suitable study partners using a matching
- algorithm Communicate via an in-built chat
  system The goal is to improve productivity,

collaboration, and learning outcomes.

1.5 References

No prior documents were used in preparing this SRS

2. Overall Description

2.1 Product Perspective

The system is a standalone web application with four modules:

1. Smart Study Planner
2. Partner Finder
3. Real-time Chat
4. Admin Panel

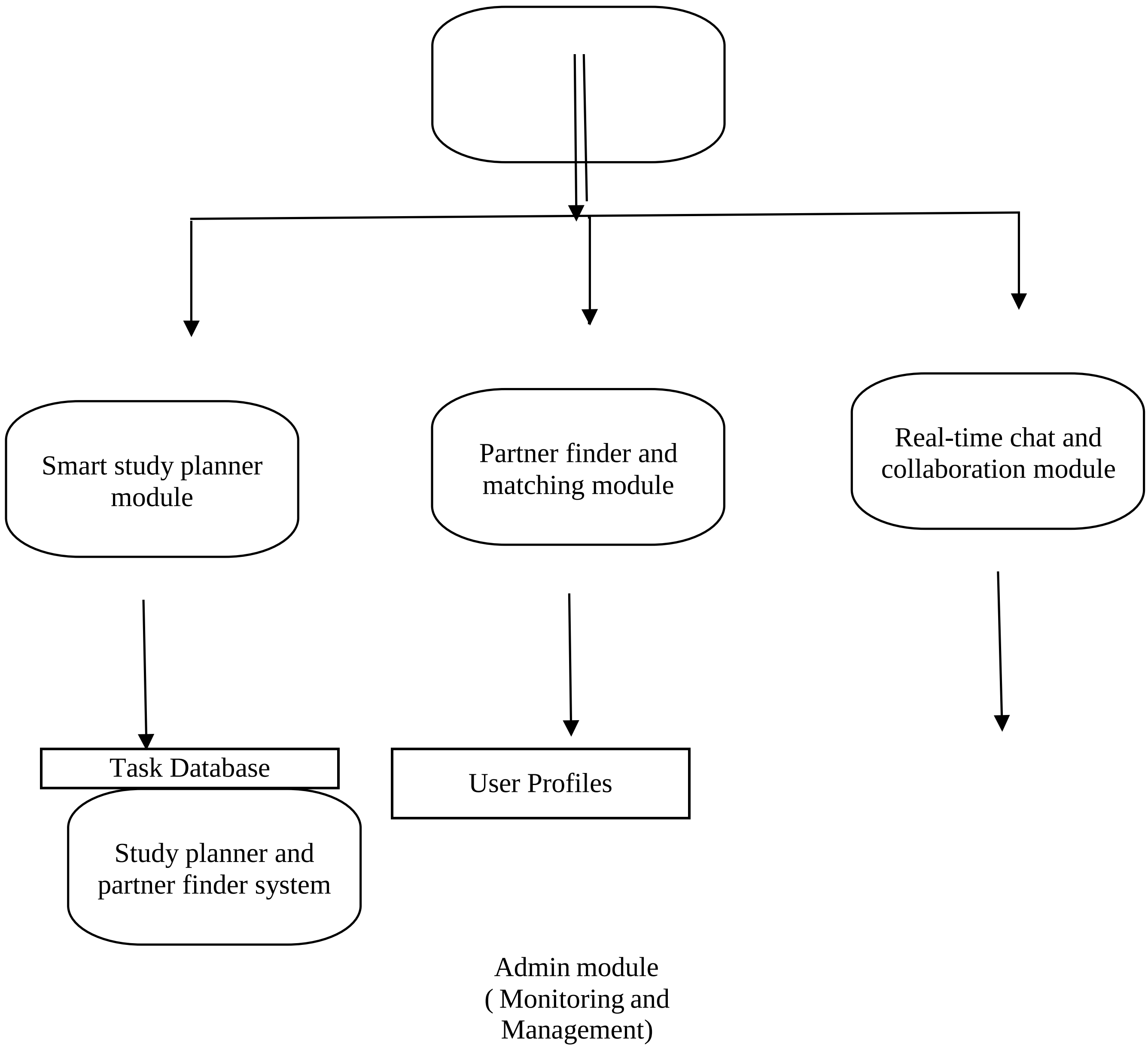
It includes a database, REST API backend, and browser-based frontend.

2.2 Product Functions

- User registration & authentication
- Creation and management of study plans
- Automatic schedule generation
- Task tracking & progress analysis
- Partner matching using multi-factor algorithm
- Chat & collaboration
- Admin monitoring & user management

Chat Database

A **conceptual diagram** showing the major modules and how they relate is included below:



2.3User Classes and Characteristics

User Type	**Characteristics**	**Access Level**
Student	Basic computer skills	Can create plans, match
		partners
Study Partner	Same as student	Collaborative tasks, chat
Admin	Technical knowledge	Full control, moderation

2.4 Operating Environment

- OS: Windows, macOS, Linux
- Browser: Chrome, Firefox, Safari
- Backend: Node.js / Django
- Database: PostgreSQL / MySQL

2.5 Design and Implementation Constraints

- Must use relational database
- Must support mobile-responsive UI
- Must support secure login (JWT or session-based) Real-
- time chat requires WebSockets

2.6 User Documentation

- Online user manual
- FAQs
- Walkthrough tutorial In-
- app tooltips

2.7 Assumptions and Dependencies

- Users have internet access
- Browser supports JavaScript
- Notifications depend on SMTP or push notification service

3.External Interface Requirements

3.1 User Interfaces

- Clean, responsive UI
- Dashboard with schedule, tasks, analytics
- Partner matching screen
- Chat window with live typing indicator

3.2 Hardware Interfaces
No special hardware required.

3.3 Software Interfaces

- Database connection
- Email server
- WebSocket server for real-time chat

3.4 Communications Interfaces

- HTTPS for secure communication
- WebSocket for chat
- SMTP for notifications

4. System Features

4.1 System Feature: Smart Study Planner

4.1.1 Description & Priority

A scheduling assistant that creates intelligent study plans.
Priority: High

4.1.2 Stimulus/Response Sequences

1. User enters subjects, deadlines, study hours.
2. System analyzes workload.
3. System generates optimized schedule.
4. User updates tasks; system updates analytics.

4.1.3 Functional Requirements

REQ-1.1: The system shall allow users to enter subjects, tasks, and deadlines.

REQ-1.2: The system shall automatically generate a study schedule using prioritization logic.

REQ-1.3: The system shall reschedule missed tasks.

REQ-1.4: The system shall display analytics, including daily hours, completion rate, and weak subjects.

REQ-1.5: The system shall send reminders for upcoming deadlines.

4.2 System Feature: Partner Finder & Matching

4.2.1 Description & Priority

Matches students based on multiple parameters.
Priority: High

4.2.2 Stimulus/Response Sequences

1. User completes profile (subjects, levels, availability).
2. System calculates match score.
3. Recommended partners are displayed.
4. User may send a collaboration request.

4.2.3 Functional Requirements

REQ-2.1: The system shall collect user preferences and study habits.
REQ-2.2: The system shall calculate match score using weighted criteria.
REQ-2.3: The system shall show partner suggestions based on match score.
REQ-2.4: The system shall notify users of new partner requests.
REQ-2.5: The system shall allow users to accept/decline partner requests.

4.3 System Feature: Real-time Chat & Collaboration

4.3.1 Description & Priority

Provides messaging and shared study space.
Priority: Medium

4.3.2 Stimulus/Response Sequences

1. Users accept match request.
2. Chat screen opens.
3. Users exchange messages in real time.
4. Users share notes, schedules, and updates.

4.3.3 Functional Requirements

REQ-3.1: The system shall support real-time messaging.
REQ-3.2: The system shall show online/offline status.
REQ-3.3: The system shall allow sharing of schedules and tasks. REQ3.4: The system shall store chat history.

4.4 System Feature: Admin Management

4.4.1 Description & Priority

Ensures system integrity and content moderation.
Priority: Medium

4.4.2 Functional Requirements

REQ-4.1: Admin can view all users.

REQ-4.2: Admin can remove inappropriate users. REQ4.3: Admin can view system statistics.

5.Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall handle 500+ concurrent users.
- Auto-scheduling algorithm shall generate a plan within 5 seconds.
- Chat messages shall appear within 1 second.

5.2 Safety Requirements

- System shall prevent data loss during power or server failure.
- Regular backups must be maintained.

5.3 Security Requirements

- All passwords must be encrypted.
- Two-factor authentication (optional).
- Chat data must be secured using encryption.

5.4 Software Quality Attributes

- **Usability:** Simple UI for students
- **Reliability:** 99% uptime
- **Maintainability:** Modular code
- **Portability:** Works on all major browsers

5.5 Business Rules

- Users must verify their email before using advanced features.
- Misuse of chat can lead to account suspension.

6.Other Requirements

- Database must support relational integrity.
- System should support future mobile app expansion.

7.Appendix A: Glossary

- **Match Score:** Numerical value representing partner compatibility
- **Task:** A study-related activity with a defined deadline
- **Analytics:** A breakdown of performance indicators

8.Appendix B: Analysis Models

- Use Case Diagrams
- Class Diagrams
- Sequence Diagrams
- Collaborative Diagrams

Detailed Use Case, Sequence, Activity, and DFD diagrams are available in a separate document.

9.Appendix C: To Be Determined List

- AI complexity level for auto-scheduling
- Push notification service provider